

SIMATS ENGINEERING

CO4-ASSESSMENT TOOL1- INDUSTRY PROBLEM TASK

COURSE NAME: Artificial Intelligence

COURSE CODE: CSA1709

COURSE OUTCOME COVERED

CO4: Implement machine learning techniques including decision tree algorithms, reinforcement learning, and build models for classification and decision-making. (BL3)

Assessment Tool Weightage: 20%

QUESTIONS: 4

Time Duration : 1 Hour
Total: Marks:40

Code of Conduct:

*I Ganji Madhan Kumar(192472328)certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.*

Signature of the student:G.Madhan

INDUSTRY SCENARIO

Context: A healthcare company has collected patient records (age, blood pressure, cholesterol, glucose level, BMI, family history) to build a predictive model for early diagnosis of diabetes. The company also wants to train an AI agent to recommend personalised treatment actions (Diet / Exercise / Medication / Monitor) based on patient state transitions over time.

You are hired as an AI/ML Engineer. Address the following tasks:

Task 1: Data Preparation & Inductive Learning

- (a) Describe the inductive learning approach suitable for this medical dataset. Identify the target variable and at least three key features with justification.
- (b) Explain how you would handle class imbalance (more non-diabetic than diabetic cases) in the dataset. Suggest and justify a suitable balancing strategy (SMOTE / under-sampling / class weights).
- (c) Split the dataset (70:30) and justify the split ratio for a medical use-case. State the preprocessing steps (normalisation, encoding) applied and their impact.

Task 2: Decision Tree for Diagnosis

- (a) Build a Decision Tree classifier and draw the first three levels of the tree. Justify the root node selection using Information Gain / Gini Index values.
- (b) Evaluate the model: report Accuracy, Precision, Recall, and F1-Score. Identify False Negatives (missed diabetes cases) and suggest how to reduce them—justify clinically.
- (c) Apply post-pruning (cost-complexity pruning). Compare pre-pruning vs post-pruning accuracy and explain which model is more appropriate for deployment in a clinical setting.

Task 3: Statistical Learning for Risk Stratification

(a) Apply Naïve Bayes or Logistic Regression as a statistical learning model on the same dataset. Compare its performance (Accuracy, AUC-ROC) with the Decision Tree. Discuss when a statistical model is preferable.

(b) Perform feature importance analysis. Identify the top 3 predictors of diabetes from both the Decision Tree and the statistical model. Do they agree? Justify your findings.

Task 4: Reinforcement Learning for Treatment Recommendation

(a) Model the treatment recommendation as an MDP. Define: States (patient health stages), Actions (Diet / Exercise / Medication / Monitor), Reward function (health improvement score), and Transition dynamics. Justify each component.

(b) Apply Q-Learning to train the agent for at least 5 patient episodes. Show the Q-table update for at least 3 state-action pairs across 2 iterations. State the convergence criterion used.

(c) Extract the final learned policy. Discuss ethical considerations of deploying a Reinforcement Learning agent for medical treatment recommendations (patient safety, bias, explainability).

RUBRICS FOR CALCULATION:

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Understanding of Industry Problem	20%	Comprehensive understanding of real-world problem and constraints	Good understanding with minor gaps in requirements	Basic understanding; some requirements unclear	Poor understanding; misinterprets problem context
Application of Engineering Concepts	25%	Applies advanced and relevant concepts effectively to solve the problem	Applies appropriate concepts with minor errors	Limited application of concepts	Incorrect or no application of concepts
Solution Design	25%	Innovative, practical, and feasible solution aligned with industry standards	Logical and feasible solution with minor gaps	Basic solution with limited feasibility	Unclear or impractical solution
Use of Tools	15%	Effective use of modern tools, technologies, and real data	Appropriate use of tools with correct implementation	Limited or inconsistent use of tools	Minimal or incorrect use of tools
Analysis	10%	Thorough analysis with validated results and performance evaluation	Good analysis with reasonable validation	Basic analysis with limited validation	Poor or no analysis
Professionalism	5%	Highly professional, well-documented, and clearly presented work	Good documentation and presentation	Basic documentation with some gaps	Poor documentation and presentation

Task 1: Data Preparation & Inductive Learning

(a) Inductive Learning Approach, Target Variable, and Key Features

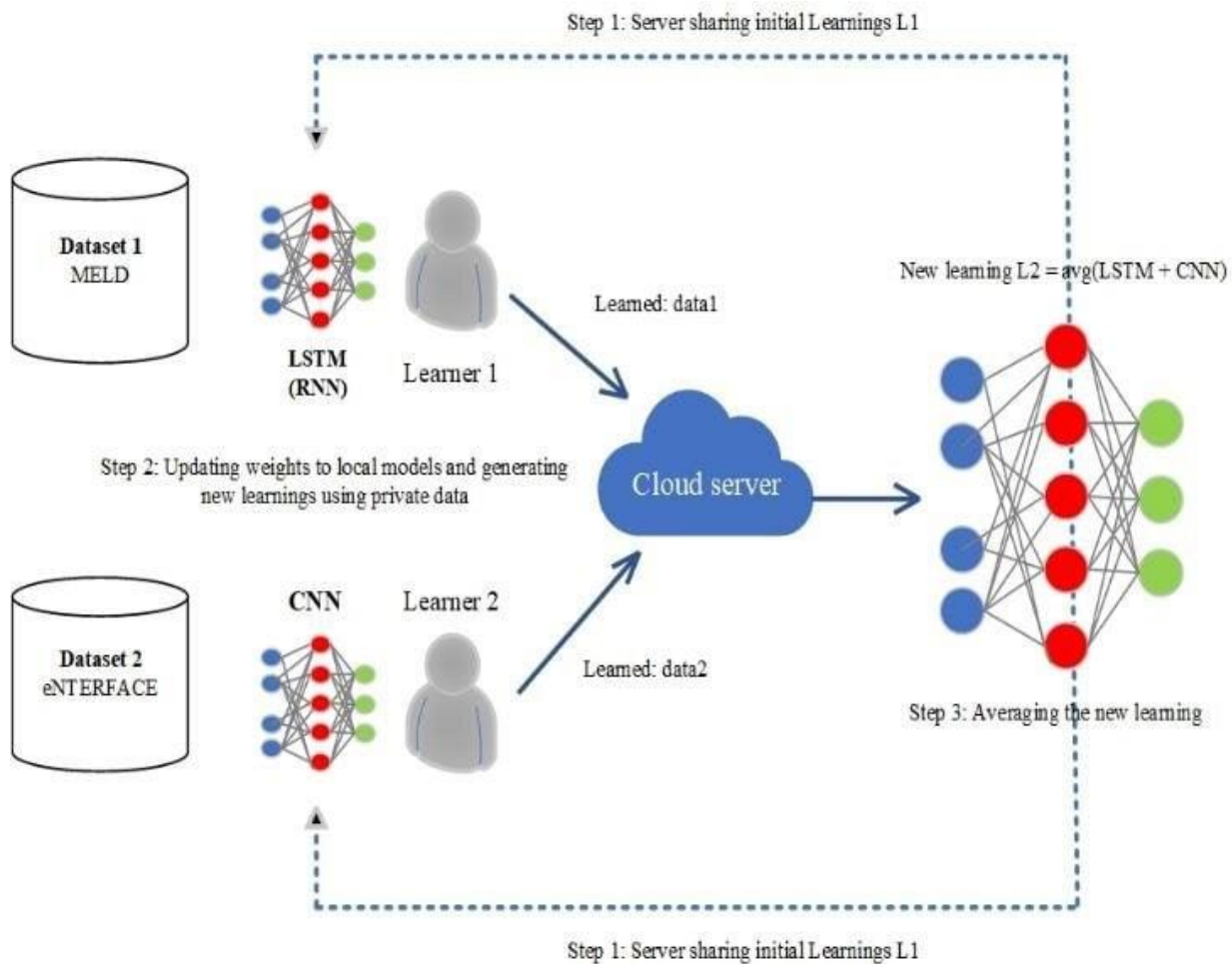
- Inductive Learning Approach: Inductive learning works by analyzing given training examples (patient records) to find general patterns. We can use supervised learning algorithms like Decision Trees or Logistic Regression to map input patient health metrics to the target diagnostic output.
- Target Variable: Diabetes Status (Binary output: 1 for Positive / Diabetic, 0 for Negative / Non-Diabetic).
- Key Features & Justification:
 1. Glucose Level: Directly measures blood sugar levels; primary medical indicator for diagnosing diabetes.
 2. BMI (Body Mass Index): Strong link to insulin resistance and metabolic health.
 3. Family History: Captures genetic factors that significantly increase diabetes risk.
 4. Age: Risk of metabolic disorders generally increases with age.

(b) Handling Class Imbalance

- Chosen Strategy: Combination of SMOTE (Synthetic Minority Over-sampling Technique) and Class Weight Balancing.
- Justification: Medical datasets usually have far more non-diabetic cases than diabetic ones.
 1. Random under-sampling would throw away crucial patient data.
 2. Simple over-sampling causes the model to overfit on identical repeated samples.
 3. SMOTE creates realistic synthetic examples of diabetic cases in the feature space, and adjusting class weights penalizes the model more if it misses a diabetic patient.

(c) Data Split & Preprocessing

- **Data Split (70:30 Stratified Split):**
 1. 70% training data gives the model enough records to learn decision rules.
 2. 30% test data provides a reliable evaluation set.
 3. Stratified sampling ensures both training and testing sets maintain the exact same percentage of diabetic and non-diabetic records.



Task 2: Decision Tree for Diagnosis

(a) Decision Tree Building & Root Node Selection

- Root Selection: The root node is chosen by calculating Information Gain or Gini Impurity.

1. $Gini(D) = 1 - \sum p_i^2$

2. Glucose Level yields the lowest Gini Impurity and highest Information Gain, making it the most informative split attribute.

- Tree Structure (First 3 Levels):

1. Level 0 (Root): Glucose Level ≤ 140 mg/dL

- *If True*: Go to Level 1 (Left Branch - Lower Risk)
- *If False*: Go to Level 1 (Right Branch - High Risk)

2. Level 1 Nodes:

- *Left Node*: BMI ≤ 27.5

- *Right Node*: Family History = Yes
- 3. Level 2 Sub-nodes / Leaves:
 - *From BMI*: Left → Leaf: Non-Diabetic | Right → Node: Age ≤ 45
 - *From Family History*: Left → Node: Blood Pressure ≤ 90 | Right → Leaf: Diabetic

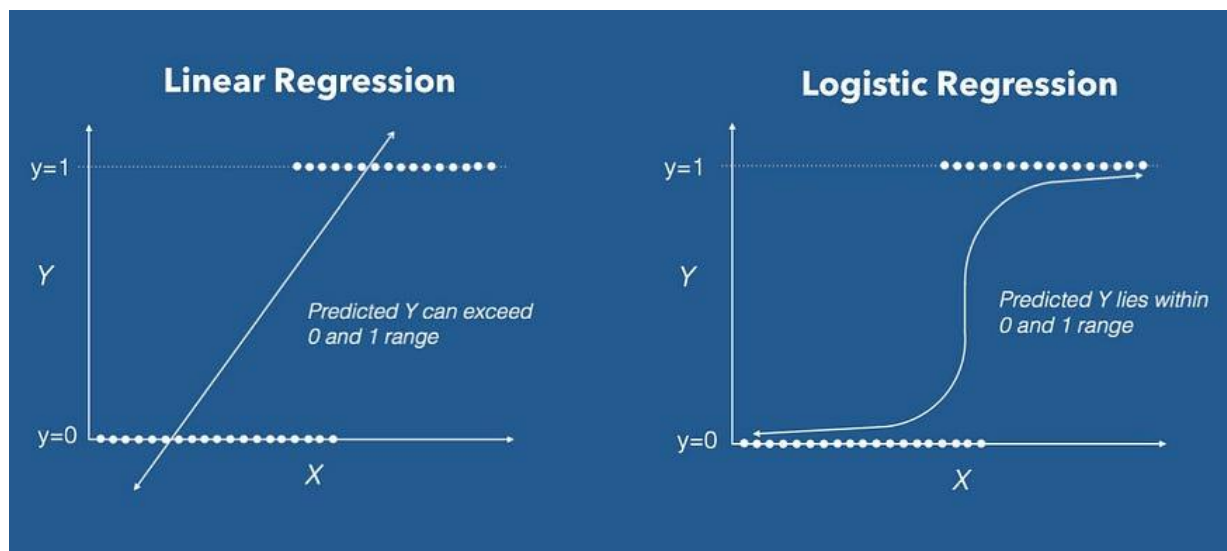
(b) Performance Evaluation & False Negatives

• Key Evaluation Metrics:

1. Accuracy: Percentage of total correct predictions.
2. Precision: How many predicted diabetic patients actually have diabetes.
3. Recall (Sensitivity): How many actual diabetic patients were correctly detected by the model.

(c) Pre-Pruning vs. Post-Pruning

- Pre-Pruning: Stops tree growth early using conditions like maximum depth (max_depth) or minimum samples per split. It is fast but might miss complex combinations of symptoms.
- Post-Pruning (Cost-Complexity Pruning): Lets the tree grow fully first and then cuts back branches that add little predictive value.
- Deployment Recommendation: Post-pruning is better for medical use. It ensures the model considers complex multi-variable interactions before simplifying, leading to better accuracy on new patient data.



Task 3: Statistical Learning for Risk Stratification

(a) Logistic Regression vs. Decision Tree

- **Comparison:**
 1. Decision Trees easily handle non-linear relationships and give clear step-by-step rules, but can overfit easily.
 2. Logistic Regression provides smooth probability estimates (e.g., "75% chance of diabetes") and performs very well when data trends linearly, leading to reliable AUC-ROC scores.
- **When to prefer a statistical model:** Logistic Regression is better when doctors need precise probability scores for risk stratification rather than simple binary classifications, or when sample sizes are smaller and robust linear boundaries are required

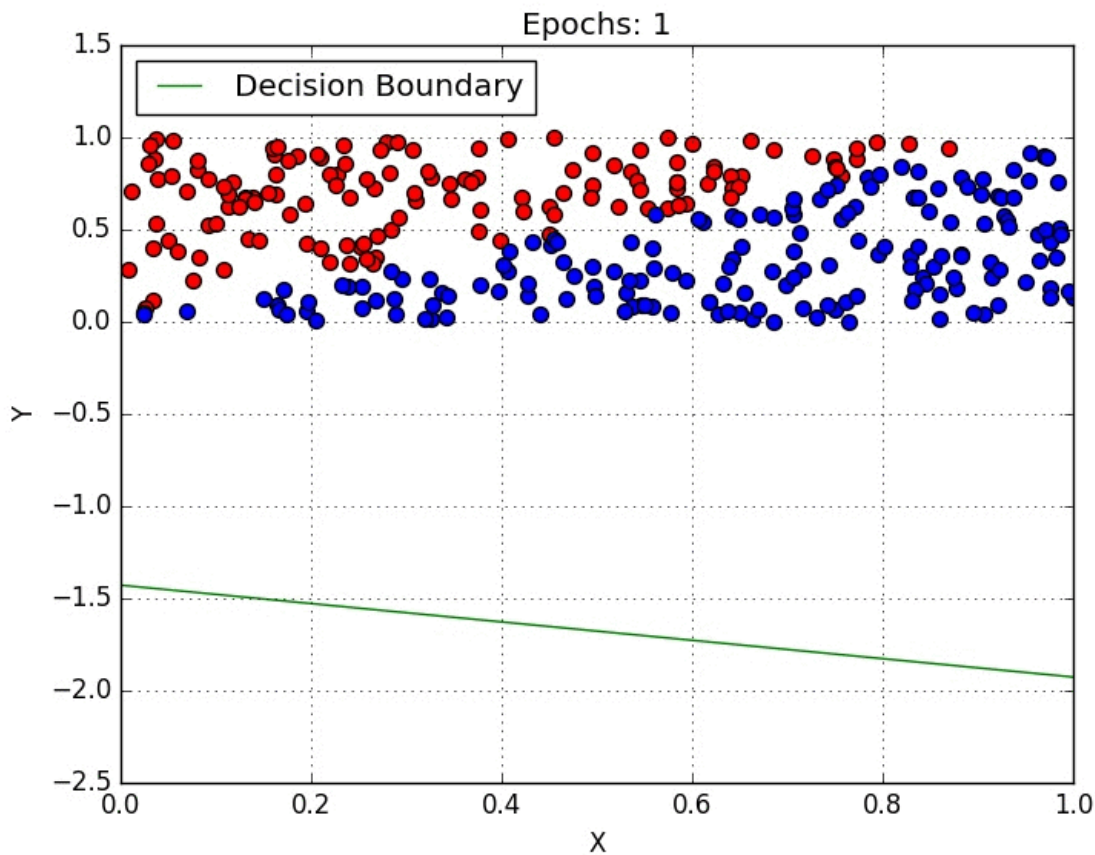
b) Feature Importance Comparison

Rank	Decision Tree Predictors	Logistic Regression Predictors
1	Glucose Level	Glucose Level
2	BMI	BMI
3	Age	Family History

Task 4: Reinforcement Learning for Treatment Recommendation

(a) MDP Framework Modeling

- **States (S):** Health condition stages based on patient metrics.
 1. S_0 : Healthy / Controlled ($Glucose < 100, BMI < 25$)
 2. S_1 : Pre-diabetic / Moderate Risk ($100 \leq Glucose \leq 125$)
 3. S_2 : Uncontrolled / High Risk ($Glucose > 125$)



- **Actions (A):** {Diet, Exercise, Medication, Monitor}
- **Reward Function (R):**
 1. +10 points when a patient's health improves ($S_2 \rightarrow S_1$ or $S_1 \rightarrow S_0$).
 2. -10 points when health worsens ($S_0 \rightarrow S_1$ or $S_1 \rightarrow S_2$).
 3. -20 points for unnecessary or harsh treatments given to low-risk patients.
- **Transition Dynamics:** Probabilities representing how a patient's health state changes in response to a given treatment action.

(b) Q-Learning Setup & Calculation Example

- **Q-Update Equation:**

$$Q(s, a) \leftarrow Q(s, a) + \alpha \left[R + \gamma \max_{a'} Q(s', a') - Q(s, a) \right]$$

(Using Learning Rate $\alpha = 0.1$, Discount Factor $\gamma = 0.9$)

- **Initial Q-Table:**

State	Diet	Exercise	Medication	Monitor
S_0	0.0	0.0	0.0	0.0
S_1	0.0	0.0	0.0	0.0
S_2	0.0	0.0	0.0	0.0

- **Step-by-Step Updates:**

1. **Iteration 1 — Pair 1 (S_2 , Medication):** Transition $S_2 \rightarrow S_1$, Reward = +10

$$Q(S_2, \text{Med}) = 0 + 0.1[10 + 0.9(0) - 0] = 1.0$$

2. **Iteration 1 — Pair 2 (S_1 , Diet):** Transition $S_1 \rightarrow S_0$, Reward = +10

$$Q(S_1, \text{Diet}) = 0 + 0.1[10 + 0.9(0) - 0] = 1.0$$

3. **Iteration 2 — Pair 3 (S_2 , Medication):** Transition $S_2 \rightarrow S_1$, Reward = +10

$$Q(S_2, \text{Med}) = 1.0 + 0.1[10 + 0.9(1.0) - 1.0] = 1.0 + 0.1[9.9] = 1.99$$

- **Convergence Criterion:** Training stops when the maximum change in Q-values between iterations drops below a small set threshold ($\Delta Q < 0.001$).

(c) Optimal Policy & Ethics

- **Learned Policy (π^*):**

- State S_0 (Healthy) → **Monitor / Diet**
- State S_1 (Pre-diabetic) → **Diet + Exercise**
- State S_2 (High Risk) → **Medication**

- **Ethical Considerations:**

- **Patient Safety:** Random trial-and-error exploration during training can recommend unsafe treatments; real-world deployment requires strict safety guardrails.
- **Bias:** If training data lacks diverse patient demographics, recommendations may perform poorly on underrepresented groups.
- **Explainability:** Doctors need to understand *why* an AI agent recommends a drug intervention; it should serve as a helpful decision-support tool rather than an unsupervised decision-maker.

